

WEDNESDAY, AUGUST 26 4:00–5:00 P.M.

Lab 4: From a policy question to a Quarto brief

One table, one figure, one finding, and one limitation

Purpose

Choose one of four international-development questions and complete its Quarto starter. Your brief should let a reader identify the sample, inspect the calculation, and distinguish a descriptive result from an unsupported claim. You will write the first interpretation before asking Codex to review the work.

Choose one track

Track	Question	Starter
Health	How is income associated with under-five mortality in 2022, and how does the pattern differ by income group?	<code>materials/lab-4/health-brief.qmd</code>
Gender	Where is girls' secondary enrollment associated with lower adolescent fertility?	<code>materials/lab-4/gender-brief.qmd</code>
Access	Which economies closed the combined electricity and internet access gap fastest from 2000 to 2022?	<code>materials/lab-4/access-brief.qmd</code>
Climate	Which economies expanded renewables, reduced carbon intensity, and ended with higher real income from 2000 to 2021?	<code>materials/lab-4/climate-brief.qmd</code>

Required brief

Your rendered HTML should contain:

1. the question and a statement of the unit, sample, period, and source;
2. one table produced from the supplied CSV;
3. one labeled figure produced from the same analytical comparison;
4. one numerical finding written in your own words;
5. one limitation or inference the evidence does not support; and
6. an AI-use note describing the statement checked, independent R evidence, and whether you accepted, revised, or rejected it.

Route through the hour

Minutes	Work	Result
00–07	Choose and plan	State the unit, variables, and function sequence.
07–20	Construct table	One traceable briefing table.
20–33	Construct figure	One labeled policy comparison.
33–40	Interpret	One finding and one limitation in your words.
40–50	Agent review	Codex reproduces the table and challenges one claim.
50–56	Independent check	Test one material agent statement in R; decide accept, revise, or reject.
56–60	Render	A standalone HTML brief.

Before coding

Write down the unit and period, the variables needed, the sequence of R functions for the table, the figure mappings, one claim the evidence could support, and one claim it could not support. The comments in the starter name the relevant functions without completing the analysis.

Codex review

Use Codex only after saving your table, figure, finding, and limitation. Replace the bracketed filename in this request:

Read my saved [QMD filename] and the CSV it imports. Work read-only. Reproduce my table from the stated sample; inspect the variables, units, and mappings in my figure; then select the most consequential numerical or interpretive statement in my Finding or Limitation. Cite the exact object and values supporting your judgment. End with accept, revise, or reject and at most one proposed revision. Do not edit files, fit a model, or make a causal claim.

Choose one material statement from the response and test it with a small, independent R calculation. In the Quarto file, record the agent's statement, your check and result, and your final decision.

Interpretation reminders

All four exercises are descriptive. The health and gender files are 2022 cross-sections. The access and climate files compare common endpoints. A change between endpoints does not establish what happened in every intervening year, and an observed association does not identify a policy effect. Gross enrollment can exceed 100 by definition. The access index gives equal weight to electricity and internet changes.

Completion

Click **Render** in RStudio. Confirm that the HTML opens, includes the table and figure, and contains no stale sentence that disagrees with the output. There is no graded submission; keep the source QMD and rendered HTML as a reusable record of your work.